

Gated Layered Workflow

A brief goes in; one checked SysML-like model comes out, layer by layer. Language models write each layer, scripts carry what one layer owes the next, and a Sysprose check holds every step before the run moves on.

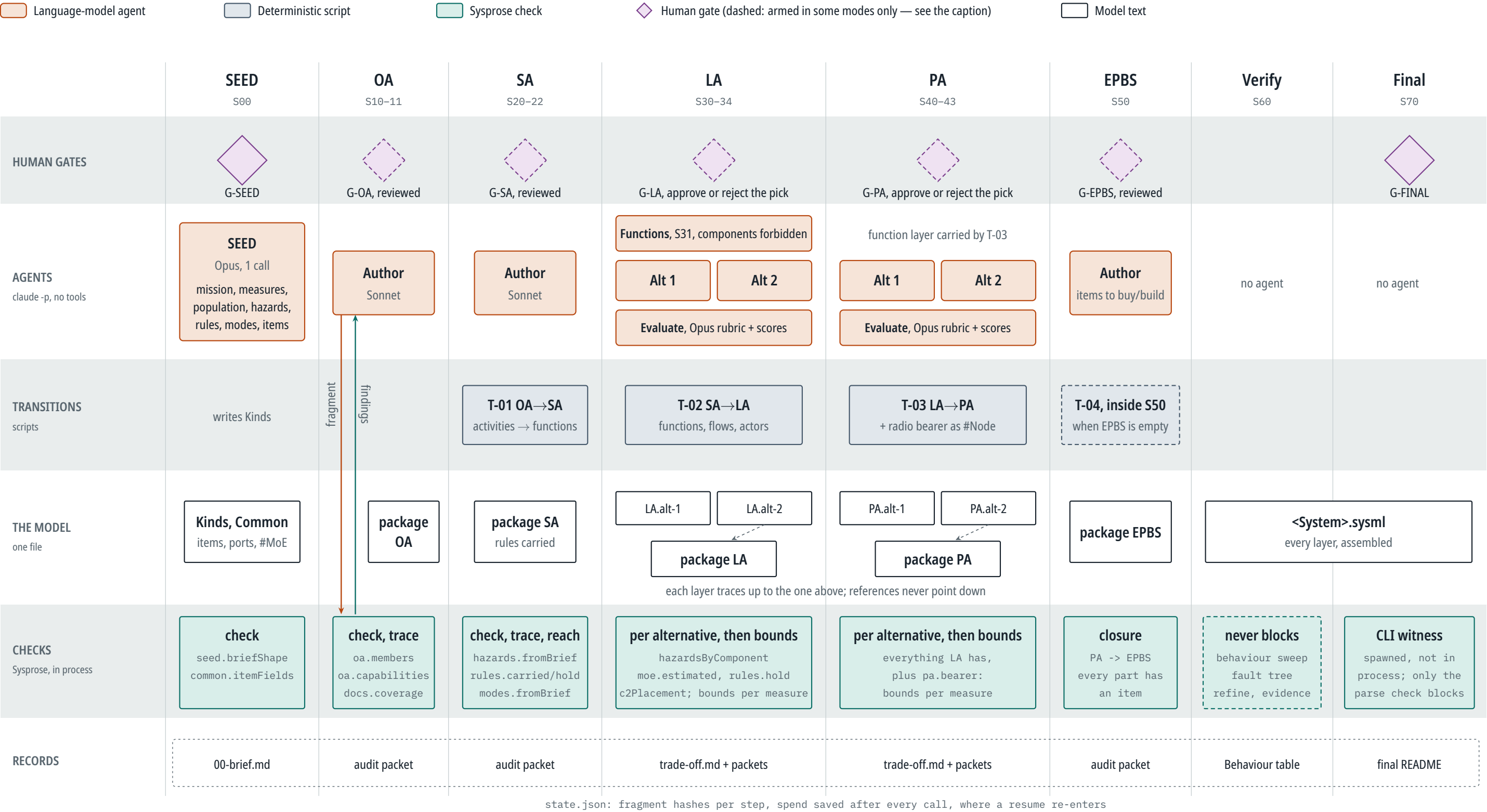


Figure 1: Six bands of responsibility against the twenty steps (S01 and S02, the optional intake lanes, are off by default and not drawn). Orange boxes are the only places a language model runs; at LA and PA it writes two architectures on the same carried function layer and an evaluator picks one. The paired arrows at OA stand for every authoring step: the fragment goes down to the checks, the findings come back up. A step that still blocks after three repair rounds *stops the run*; it opens its gate first only where a gate exists (S00, S10, S21, S33, S42, S50, S70 — the transitions and the two alternatives steps have none), and approving that gate records the decision and stops too. The way on is an edit and a resume. Solid gates are armed in every mode, but --unattended takes them automatically: in v7 both were.

One alternative-authoring step, up close

S32 and S41, where the model writes one of two architectures. The model never touches a file: what it returns is composed onto the fixed function layer, assembled with the layers above, and checked before anything is kept. A plain author — S10, S21, S31, S50 — runs the same loop without the Compose stage.

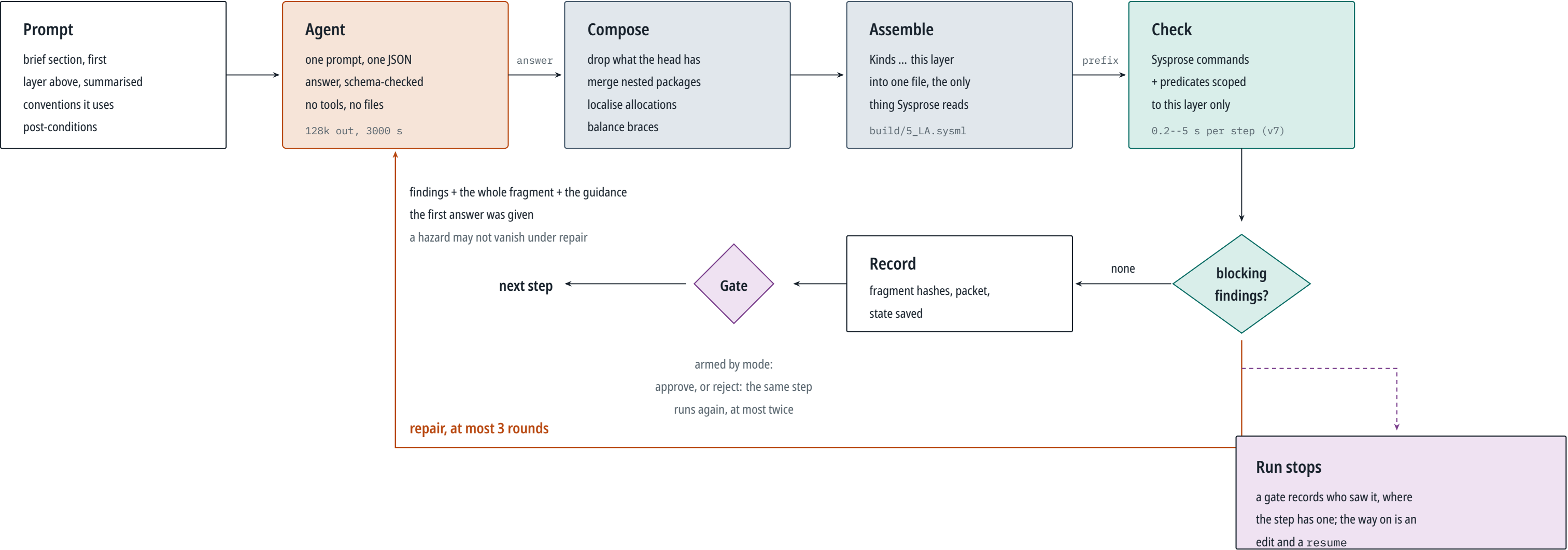


Figure 2: One alternative-authoring step (S32, S41); the model writes, it never edits files. Compose is this step's own: a plain author (S10, S21, S31, S50) has its carried trace and allocate lines restored and its answer preflighted instead. A diagnostic anchored in a layer above is reported, not sent back — the author cannot change that layer — and travels on as a note. A blocking finding goes back with the fragment it came from, at most three times; after that the run stops. Check times: 0.2–5 s per step's checks in v7.

Two architectures, one choice

At LA and PA the functions are fixed and shared, so only the split of responsibilities differs. The two alternatives are built to disagree on where coordination and command live, and the score says what that costs.

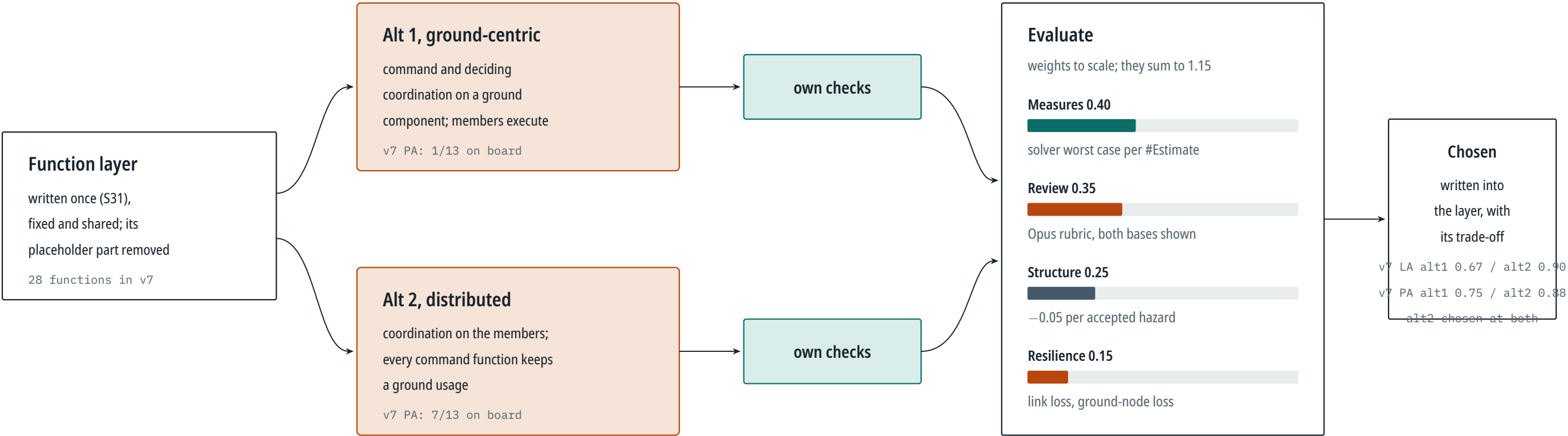


Figure 3: A measure decides only when some alternative misses it. Each estimate is the architecture’s own claim, bounded for its worst case; totals run to 1.15, not 1, because the four weights are not normalised. The final audit lists the targets every alternative met: in v7, 12 of 14. Without a population the resilience criteria are not judged and every alternative takes a neutral 0.5; the ground-centric/distributed split is a population rule, and if one alternative fails its checks the other is taken with no comparison.